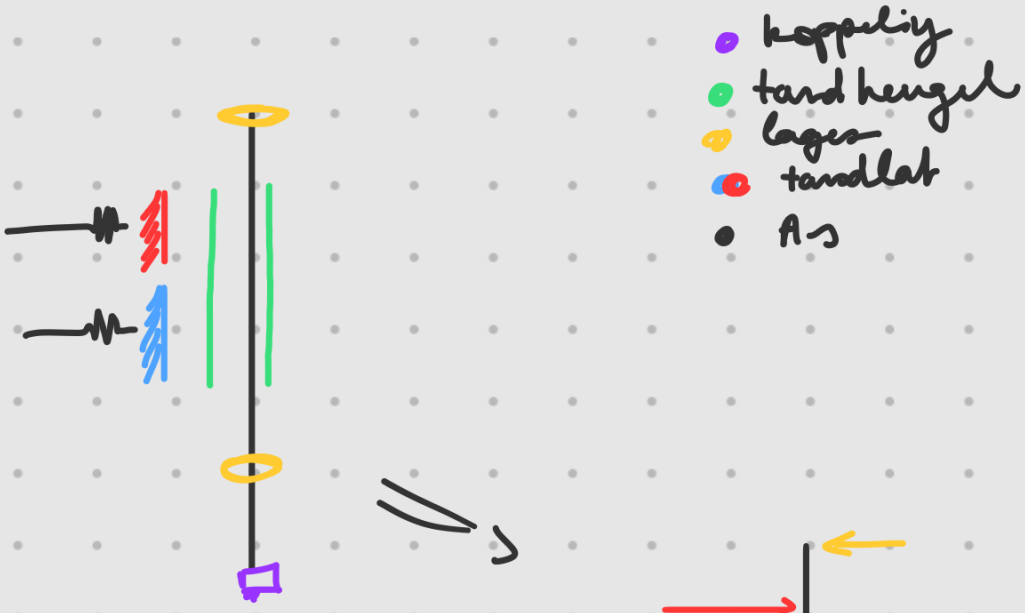
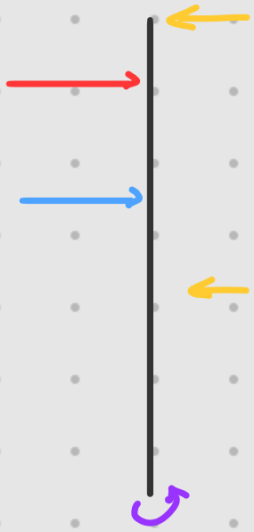




geschatte massa robot: $\boxed{200\,kg}$

geschatte ø tandwiel: 0,5° aanwaarheid

⇝ < 1 mm

$$\text{omtrek} = \frac{2\pi r \cdot 0{,}5}{360} < 1\,mm$$

$$r < 114{,}6\,mm$$

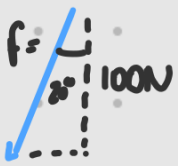
$$ø < 229{,}2\,mm$$

↳ aanname op 220 mm ø

max versnelling 0,5 m/s² voor 2 wan systemen

⇝ $\left(F = M \cdot a = 200\,kg \cdot 0{,}5\,m/s^2 = 100\,N \right.$ ← wrijving en inefficiëntie $\left. \right)$

k = 1,5 = 150 N
(bw)

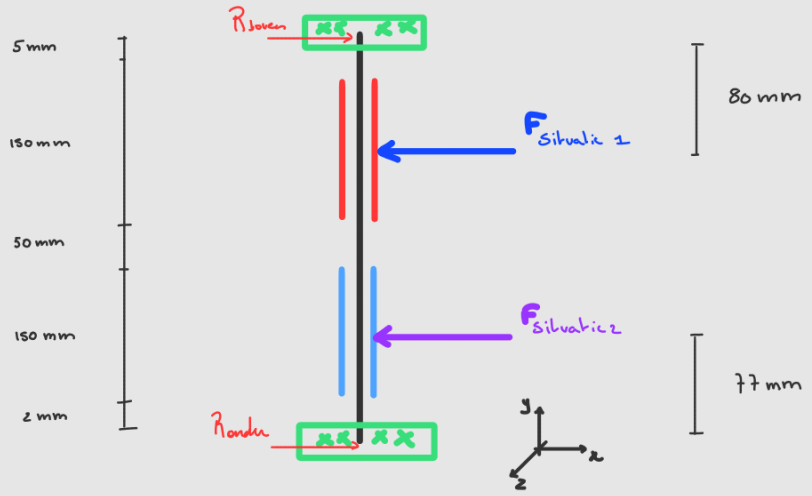


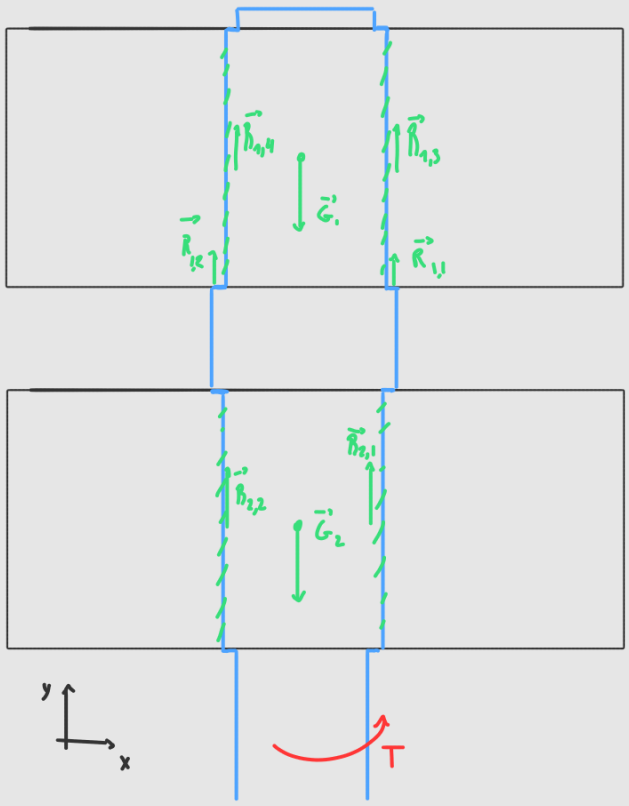
$\cos(20°) \cdot F = 100\,N$ ⇝ $F = 106{,}41\,N$

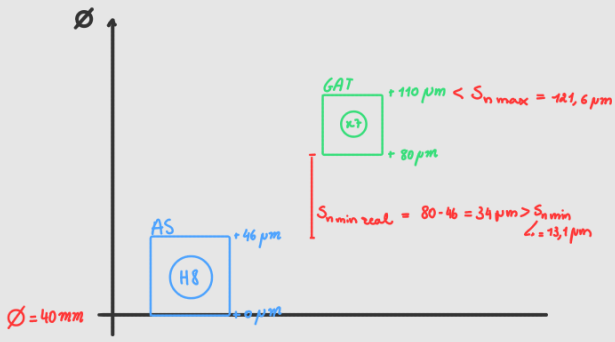
$\cdot 1{,}5 = 159{,}63\,N$

$$T = F \cdot r = 159{,}63\,N \cdot 0{,}11\,m = 17{,}56\,Nm$$

Eerste volgende stap is tandwiel buiten en klaarmaken berekeningen voorafgaan de zeef kan uitgewerkt worden




$\vec{R}_{1,4}$
$\vec{R}_{1,3}$
$\vec{G}_1$
$\vec{R}_{1,2}$
$\vec{R}_{1,1}$
$\vec{R}_{2,1}$
$\vec{R}_{2,2}$
$\vec{G}_2$
y
x
T


Ø
GAT
x7
+110 µm < $S_{n\,max}$ = 121,6 µm
+80 µm
$S_{n\,min\,real}$ = 80 - 46 = 34 µm > $S_{n\,min}$ = 13,1 µm
AS
H8
+46 µm
+0 µm
Ø = 40 mm